

Smart Fire Detection and Automatic Suppression System for Electrical Distribution Boards

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1. Introduction

The electrical distribution panels are the main points of the electrical system, where faults like overheating, insulation failure, and loose connections may lead to fire hazards. Conventional protection devices like MCBs and RCCBs are used to identify electrical faults, but they cannot identify the signs of fire like smoke, flame, or temperature.

This project proposes a fire protection system for distribution panels (mainly larger scale panels) using smoke, flame, and temperature sensors to enable early detection of fire conditions. The system continuously monitors the conditions of the electrical panels. In case of fire, the alarm signals are sent, and the exhaust fan operates based on the temperature levels. However, the exhaust fan operation is restricted when there are signs of smoke or flames, as it may worsen the fire condition. In critical conditions, the fire suppression system is used to control the fire.

In the proposed system, the smoke sensor has also been developed as a part of the project.

1. Objectives

- To design and develop a fire protection system for electrical distribution panels capable of detecting early fire conditions using temperature, smoke, and flame sensors.
- Continuous monitoring of internal panel conditions and generating timely alerts through an alarm system when abnormal conditions are detected.
- To develop a control mechanism for an exhaust fan that operates based on temperature levels to reduce heat buildup while preventing operation during fire conditions involving smoke or flame.
- To design and integrate an automatic fire suppression system that activates under extreme conditions to control or extinguish fire within the panel.
- To design and implement a smoke sensor as part of the system and integrate it with other sensing and control components.
- To evaluate the performance and response of the system under different faults and fire scenarios.

2. Project Scope

In Scope :-

- detection of fire conditions using temperature, smoke and flame sensors.
- fabrication of a self built smoke sensor and its integration with the system.
- continuous monitoring of electrical distribution panel conditions using LabVIEW & NI DAQ.
- control of actuators (buzzer, cooling fan, fire suppression system) based on sensor inputs through the LabVIEW.
- sending alerts to users through ESP32 based mobile notification system
- automatic activation of alarm (buzzer) under abnormal or fire conditions & temperature based control of cooling fan operation.

3. Literature review

This looks at recent developments in IoT-based fire detection systems. The main topics covered are how using multiple sensors can improve accuracy and how to detect fires early.

Several studies have worked on creating fire detection systems that use different types of sensors to get better results. Bimansyah et al. [1] developed an IoT-based system for electric panels that uses temperature and smoke sensors and found that it could monitor conditions effectively in real time. A typical fire alarm system includes sensors that detect smoke, heat, and flames. According to Ali et al. [2], combining these sensors improves accuracy because each one detects a different sign of fire.

In addition to smoke and heat detection, flame sensors are also commonly used. Cheng and Zhang [3] showed that combining data from multiple sensors, including flame detectors, can make the system more reliable and help reduce false alarms.

Smoke detection is often considered a good way to catch fires early. Rao [4] explained that smoke usually appears before flames, which allows for earlier warning. However, temperature sensors are also important because heat can be detected even when smoke is not yet present. Using all three types of sensors—flame, smoke, and temperature—together can provide a more complete picture of fire conditions.

Using a combination of sensors can improve fire detection accuracy and help identify fires early [1]. Most studies agree that no single sensor is enough on its own. By combining flame, smoke, and temperature sensors, a fire detection system can respond more quickly and reliably.

4. Methodology

The project begins with analysis of the requirements, including the safe temperature range inside the distribution board, smoke and flame detection sensitivity, and the optimal placement of sensors. This analysis ensures that the system can detect early fire hazards accurately and reliably while maintaining safety during demonstration.

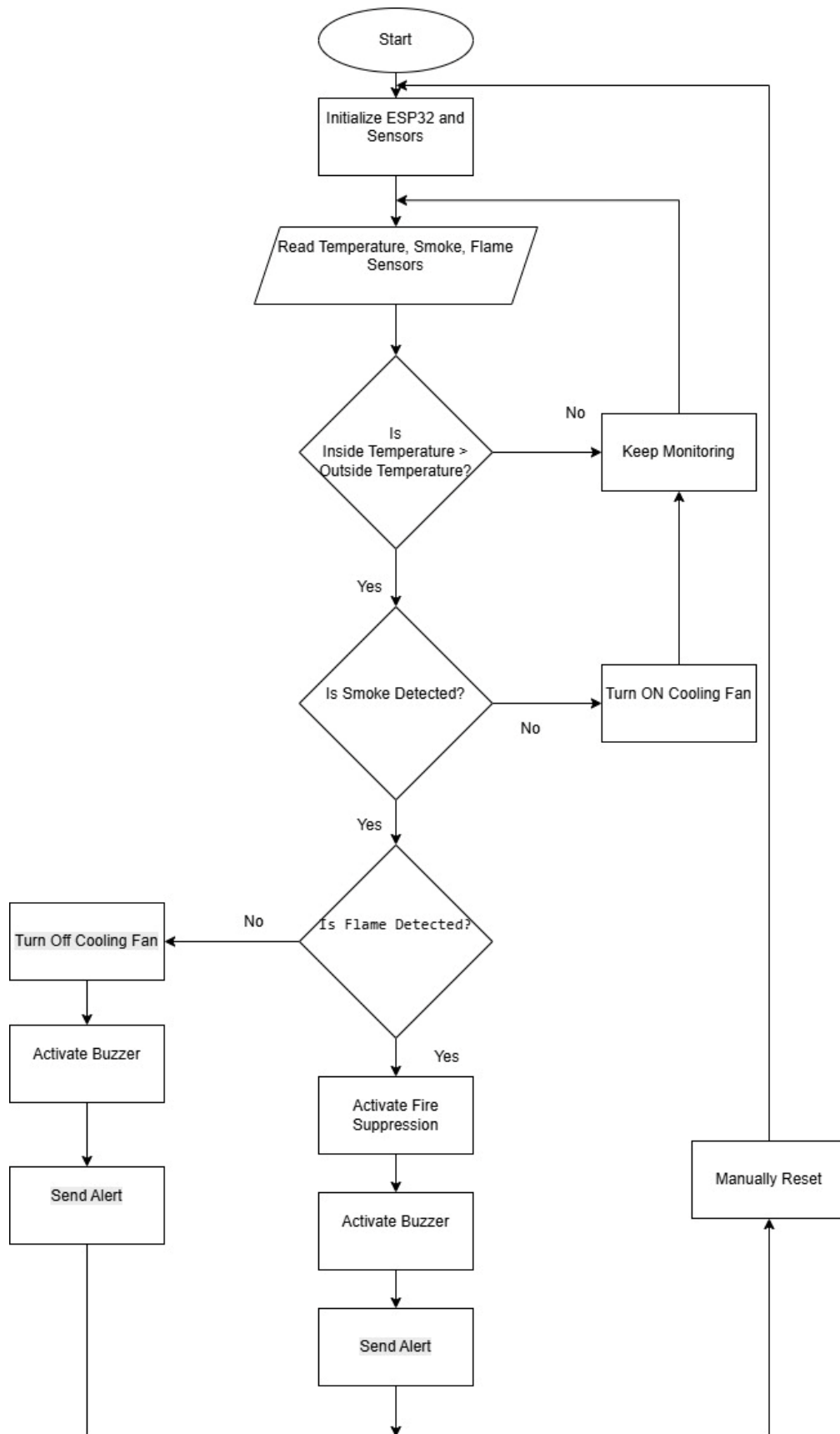
Next, the components required for the system are selected including the temperature sensor, smoke sensor, flame sensor, ESP32 microcontroller, cooling fan, buzzer. We are going to implement a prototype fire suppression mechanism system to simulate aerosol discharge. This approach ensures safe demonstration while representing the functionality of an automatic suppression system. The choice of each component is based on functionality, compatibility for the project.

In the system design and implementation phase, the sensors are connected to the DAQ card, which interfaces with a computer for real-time monitoring and decision-making. The ESP32 is used to send alerts to a mobile device. The decision logic is implemented based on predefined thresholds for temperature, smoke, and flame, which control the activation of the cooling fan, buzzer, and fire suppression system. All components are mounted and connected carefully, using proper wiring and assembly techniques to ensure safety and reliability.

Finally, the system is tested and optimized. Testing involves simulating high temperature, smoke, and flame conditions to verify that the system responds correctly, including triggering alarms, activating the fan, operating the fire suppression prototype, and sending mobile notifications.

Estimated Budget

Item	Component	Quantity	Price (LKR)
1	ESP32	1	1500
2	DS18B20 Temperature Sensor	2	400
3	MQ-2 Gas and Smoke Sensor	1	400
4	KY-026 Flame Sensor Module	1	200
5	Active Buzzer Module	1	200
6	Cooling Fan (12V DC)	1	300
7	Relay Module	1	500
8	For fire suppression prototype	1	600
9	Other small components		1000
	Total		5100



References

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